

2. Peter, Claire and David compared the hardness of three samples of water, **A**, **B** and **C**. They added soap solution to each sample until it gave a lather on shaking. They carried out a fair test.

(a) Give **two** ways they made it a fair test.

[2]

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(b) Their results are shown in the table.

| Water sample | Volume of soap solution needed to give a lather (cm <sup>3</sup> ) |        |       |       |
|--------------|--------------------------------------------------------------------|--------|-------|-------|
|              | Peter                                                              | Claire | David | Mean  |
| <b>A</b>     | 18                                                                 | 19     | 17    | ..... |
| <b>B</b>     | 8                                                                  | 10     | 9     | 9     |
| <b>C</b>     | 1                                                                  | 1      | 1     | 1     |

- (i) Calculate the mean volume of soap solution needed for sample **A** to give a lather.  
**Show your working.**

[2]

Mean = ..... cm<sup>3</sup>

- (ii) Which of the three water samples is the **hardest**? Give a reason for your answer.

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(c) The box shows some advantages and some disadvantages of hard water.

|                                    |                                    |
|------------------------------------|------------------------------------|
| <b>strengthens bones and teeth</b> | <b>forms limescale when heated</b> |
| <b>wastes soap</b>                 | <b>reduces heart disease</b>       |
| <b>forms a scum with soap</b>      | <b>tastes better</b>               |

Choose **all** the disadvantages of hard water from the box.

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(d) Choose from the box an ion that causes hardness in water.

[1]

|                      |                       |                        |                        |
|----------------------|-----------------------|------------------------|------------------------|
| <b>K<sup>+</sup></b> | <b>Na<sup>+</sup></b> | <b>Ca<sup>2+</sup></b> | <b>Cu<sup>2+</sup></b> |
|----------------------|-----------------------|------------------------|------------------------|

Ion .....

